3. Develop a menu driven Program in C for the following operations on STACK of Integers (Array Implementation of Stack with maximum size MAX >15). a. Push an Element on to Stack. b. Pop an Element from Stack. c. Demonstrate Overflow and Underflow situations on Stack. d. Display the status of Stack at any given point of time. e. Exit. Support structured program with appropriate functions for each of the above operations.

```
#include <stdio.h>
#include <stdlib.h>
#define SIZE 5
int s[SIZE], top = -1;
void push()
{
   int item;
   if (top == SIZE - 1)
   {
      printf("Stack is Full or Overflow\n");
      return;
   }
   printf("Enter an item\n");
   scanf("%d", &item);
   s[++top] = item;
}
void pop()
{
   if (top == -1)
   {
      printf("Stack is Empty or Underflow\n");
      return;
   }
   printf("Deleted item = %d\n", s[top--]);
}
void display()
{
   int i;
   if (top == -1)
   {
      printf("Stack is Empty or Underflow\n");
      return;
   }
```

```
    printf("Contents of the stack are:\n");
    for (i = top; i >= 0; i--)
    {
        printf("%d\t", s[i]);
    }
    printf("\n");
}
int main()
{
    int ch;
    while (1)
    {
        printf("\n1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice\n");
        scanf("%d", &ch);
        switch (ch)
        {
            case 1:
                push();
                break;
            case 2:
                pop();
                break;
            case 3:
                display();
                break;
            case 4:
                exit(0);
            default:
                printf("Invalid Choice\n");
        }
    }
    return 0;
```

```
}
```